Relative Covariance

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Note: *AI was heavily used as a tool for making this document. My general workflow is to discuss with AI the direction that I want to take the writeup in and then I ask it to write a first draft. I then read through the draft until I feel confident I understand it, and then typically I rewrite the document myself, borrowing certain derivations and a few sentences from the AI's initial draft. The final result is a hodge-podge of contirbutions from the AI and I.*

1 Measuring relatedness

1.1 Information and notation

1.1.1 Kinds of information

1.1.2 How the kinds of information relate

1.2 Population baseline

1.2.1 Allele frequency and standardization

1.2.2 Departures from the baseline

1.3 The fundamental processes

1.3.1 Meiosis

1.3.2 Mating

1.4 Pedigrees, paths and path diagrams

1.4.1 Path diagrams

1.4.2 Mating and meiosis graphs

1.4.3 Mating sequences

1.4.4 Lineal sequences

1.4.5 Paths between relatives

1.4.6 Path sufficiency

1.5 Measures of relatedness

1.5.1 Identity by descent

1.5.2 Realized relatedness

1.5.3 Allelic sharing

1.5.4 Founders

2 Additive genetics under random mating

2.1 The phenotype model